

61. If  $\gcd(m, n) = 1$  and  $1^2 - 2^2 + 3^2 - 4^2 + \dots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$  then  $m^2 - n^2$  is equal to:

यदि  $\gcd(m, n) = 1$  है तथा  $1^2 - 2^2 + 3^2 - 4^2 + \dots + (2021)^2 - (2022)^2 + (2023)^2 = 1012m^2n$  है, तो  $m^2 - n^2$  बराबर है:

- (A) 220 (B) 200 (C) 240 (D) 180

**Ans.** C

**Sol.**  $1^2 - 2^2 + 3^2 - 4^2 + \dots + (2021)^2 - (2022)^2 + (2023)^2 = 1012$

$$m^2n = (1-2)(1+2) + (3-4)(3+4) + \dots + (2021-2022)(2021+2022) + (2023)^2$$

$$= (-1)(1+2+3+4+\dots+2022) + (2023)^2$$

$$= (-1) \cdot \frac{(2022)(2023)}{2} + (2023)^2$$

$$= 2023(2023 - 1011) = 2023 \times 1012$$

$$m^2n = 2023 = 17^2 \cdot 7$$

$$m = 17, n = 7$$

$$m^2 - n^2 = 17^2 - 7^2 = 240$$

62. Let  $a_n$  be the  $n^{\text{th}}$  term of the series  $5 + 8 + 14 + 23 + 35 + 50 + \dots$  and  $S_n = \sum_{k=1}^n a_k$ . Then  $S_{30} - a_{40}$  is equal to:

माना श्रेणी  $5 + 8 + 14 + 23 + 35 + 50 + \dots$  का  $n^{\text{वाँ}}$  पद  $a_n$  है तथा  $S_n = \sum_{k=1}^n a_k$  तो  $S_{30} - a_{40}$  बराबर है :

- (A) 11260 (B) 11310 (C) 11280 (D) 11290

**Ans.** D

**Sol.**  $S_n = 5 + 8 + 14 + 23 + 35 + \dots + a_{n-1} + a_n$

$$S_n = 5 + 8 + 14 + 23 + \dots + a_{n-1} + a_n$$

$$a_n = 5 + 3 + 6 + 9 + 12 + \dots$$

$$a_n = 5 + \frac{n-1}{2} [6 + (n-2)3]$$

$$a_n = 5 + \left( \frac{n-1}{2} \right) (3n) = \frac{3}{2}n(n-1) + 5$$

$$a_n = \frac{3n^2}{2} - \frac{3n}{2} + 5$$

$$a_{40} = 3 \times \frac{40^2}{2} - 3 \times \frac{40}{2} + 5$$

$$= 2400 - 60 + 5 = 2345$$

$$S_{30} = \frac{3}{2} \times \frac{30 \times 31 \times 61}{6} - \frac{3}{2} \frac{30 \times 31}{2} + 5 \times 30$$

$$= 13635$$

$$S_{30} - a_{40} = 11290$$

- 63.** Let the first term  $a$  and the common ratio  $r$  of a geometric progression be positive integers. If the sum of squares of its first three terms is 33033, then the sum of these three terms is equal to :

माना एक गुणोत्तर श्रेणी के प्रथम पद  $a$  तथा सार्व अनुपात  $r$  धनात्म पूर्णांक है। यदि इसके प्रथम तीन पदों के वर्गों का योग 33033 है, तो इन तीन पदों का योग है :

- (A) 231 (B) 220 (C) 210 (D) 241

**Ans.** A

**Sol.**  $a^2 + a^2 r^2 + a^2 r^4 = 33033$

$$a^2 (1 + r^2 + r^4) = 33033$$

$$a^2 (1 + r^2 + r^4) = 3 \times 7 \times (11)^2 \times 13$$

$$\Rightarrow a^2 = (11)^2$$

$$a = 11$$

$$\Rightarrow 1 + r^2 + r^4 = 273$$

$$r^4 + r^2 - 272 = 0$$

$$\Rightarrow r^2 = 16$$

$$r = 4$$

$$a = 11$$

$$ar = 44$$

$$ar^2 = 176$$

$$a + ar + ar^2 = 231$$

- 64.** Let  $x_1, x_2, \dots, x_{100}$  be in an arithmetic progression, with  $x_1 = 2$  and their mean equal to 200. If

$y_i = i(x_i - i), 1 \leq i \leq 100$ , then the mean of  $y_1, y_2, \dots, y_{100}$  is :

माना  $x_1, x_2, \dots, x_{100}$  एक समांतर श्रेणी में हैं, जिनका माध्य 200 है तथा  $x_1 = 2$  है। यदि  $y_i = i(x_i - i), 1 \leq i \leq 100$  हैं, तो

$y_1, y_2, \dots, y_{100}$  का माध्य है :

- (A) 10051.50 (B) 10049.50 (C) 10101.50 (D) 10100

**Ans.** B

**Sol.**  $2 + (2 + d) + \dots + (2 + 99d) = 20000$

$$\Rightarrow 200 + \frac{99}{2} \times 100d = 20000$$

$$\Rightarrow 99d = 396$$

$$d = 4$$

$$y_1 + y_2 + \dots + y_{100} = \sum_{i=1}^{100} i(2 + (i-1)4 - i) = \sum_{i=1}^{100} (3i^2 - 2i) = \frac{3 \cdot 100 \cdot 101 \cdot 201}{6} - 100 \cdot 101$$

$$= 10100 \left( \frac{201}{2} - 1 \right) = (10100) \cdot \left( \frac{199}{2} \right)$$

$$\text{So mean} = \frac{(10100)(99.5)}{100} = 10049.5$$

65. Let a, b, c and d be positive real numbers such that  $a + b + c + d = 11$ . If the maximum value of  $a^5 b^3 c^2 d$  is  $3750\beta$ , then the value of  $\beta$  is :

माना a, b, c तथा d धनात्मक वास्तविक संख्याएँ हैं तथा  $a + b + c + d = 11$  है। यदि  $a^5 b^3 c^2 d$  का उच्चतम मान  $3750\beta$  है, तो  $\beta$  का मान है :

(A) 108

(B) 110

(C) 90

(D) 55

Ans. C

Sol. Given  $a + b + c + d = 11$  ( $a, b, c, d > 0$ )

$$(a^5 b^3 c^2 d)_{\max} = ?$$

Let assume Numbers –

$$\frac{a}{5}, \frac{a}{5}, \frac{a}{5}, \frac{a}{5}, \frac{a}{5}, \frac{b}{3}, \frac{b}{3}, \frac{b}{3}, \frac{c}{2}, \frac{c}{2},$$

We know A.M.  $\geq$  G.M.

$$\frac{\frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{a}{5} + \frac{b}{3} + \frac{b}{3} + \frac{b}{3} + \frac{c}{2} + \frac{c}{2} + d}{11} \geq \left( \frac{a^5 b^3 c^2 d}{5^5 \cdot 3^3 \cdot 2^2 \cdot 1} \right)^{\frac{1}{11}}$$

$$\frac{11}{11} \geq \left( \frac{a^5 b^3 c^2 d}{5^5 \cdot 3^3 \cdot 2^2 \cdot 1} \right)^{\frac{1}{11}}$$

$$a^5 \cdot b^3 \cdot c^2 \cdot d \leq 5^5 \cdot 3^3 \cdot 2^2$$

$$\max(a^5 b^3 c^2 d) = 5^5 \cdot 3^3 \cdot 2^2 = 337500$$

$$= 90 \times 3750 = \beta \times 3750$$

$$\beta = 90$$

66. Let  $\langle a_n \rangle$  be a sequence such that  $a_1 + a_2 + \dots + a_n = \frac{n^2 + 3n}{(n+1)(n+2)}$ . If  $28 \sum_{k=1}^{10} \frac{1}{a_k} = p_1 p_2 p_3 \dots p_m$ , where  $p_1, p_2, \dots, p_m$  are the first  $m$  prime numbers, then  $m$  equal to :

माना  $\langle a_n \rangle$  एक अनुक्रम है जिसके लिए  $a_1 + a_2 + \dots + a_n = \frac{n^2 + 3n}{(n+1)(n+2)}$  है। यदि  $28 \sum_{k=1}^{10} \frac{1}{a_k} = p_1 p_2 p_3 \dots p_m$  है, जहाँ

$p_1, p_2, \dots, p_m$  प्रथम  $m$  अभाज्य संख्या है तो  $m$  बराबर है :

- (A) 5 (B) 7 (C) 6 (D) 8

**Ans.** C

**Sol.** 
$$a_n = \frac{n^2 + 3n}{(n+1)(n+2)} - \frac{(n-1)^2 + 3(n-1)}{n(n+1)}$$

$$a_n = \frac{n^3 + 3n^2 - (n+2)(n^2 + n - 2)}{n(n+1)(n+2)}$$

$$a_n = \frac{4}{n(n+1)(n+2)}$$

$$\sum_{k=1}^{10} \frac{1}{a_k} = \frac{1}{4} \sum_{k=1}^{10} (k(k+1)(k+2))$$

$$\sum_{k=1}^{10} \frac{1}{a_k} = \frac{1}{16} \sum_{k=1}^{10} (k(k+1)(k+2)(k+3) - (k-1)k(k+1)(k+2))$$

$$\sum_{k=1}^{10} \frac{1}{a_k} = \frac{1}{16} (10 \cdot 11 \cdot 12 \cdot 13)$$

$$28 \sum_{k=1}^{10} \frac{1}{a_k} = \frac{28}{16} \cdot 2 \cdot 5 \cdot 11 \cdot 3 \cdot 4 \cdot 13 \Rightarrow m = 6$$

67. Let  $s_1, s_2, s_3, \dots, s_{10}$  respectively be the sum to 12 terms of 10 A.P. s whose first terms are 1, 2, 3, ..., 10 and the common differences are 1, 3, 5, ..., 19 respectively. Then  $\sum_{i=1}^{10} s_i$  is equal to :

माना 10 A.P., जिनके प्रथम पद 1, 2, 3, ..., 10 तथा सार्व अंतर क्रमशः 1, 3, 5, ..., 19 हैं, के 12 पदों का योग क्रमशः  $s_1, s_2,$

$s_3, \dots, s_{10}$  है। तो  $\sum_{i=1}^{10} s_i$  बराबर है :

- (A) 7380 (B) 7360 (C) 7260 (D) 7220

**Ans.** C

**Sol.**  $1 \leq r \leq 10$

$$S_r = \frac{12}{2} [2r + (12-1)(2r-1)] = 6(2r + 22r - 11)$$

$$= 6(24r - 11)$$

$$\sum_{r=1}^{10} S_r = 6 \sum_{r=1}^{10} (24r - 11) = 6 \times \frac{10}{2} [24 - 11 + 240 - 11]$$

$$= 30 [242] = 7260$$

**68.** Let  $A_1$  and  $A_2$  be two arithmetic means and  $G_1, G_2, G_3$  be three geometric means of two distinct positive numbers.

Then  $G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$  is equal to :

यदि दो धनात्मक विभिन्न संख्याओं के मध्य दो समान्तर माध्य क्रमशः  $A_1, A_2$  तथा तीन गुणोत्तर माध्य  $G_1, G_2, G_3$  हैं, तो

$G_1^4 + G_2^4 + G_3^4 + G_1^2 G_3^2$  का मान बराबर होगा :

(A)  $(A_1 + A_2)G_1^2 G_3^2$  (B)  $2(A_1 + A_2)G_1 G_3$  (C)  $(A_1 + A_2)^2 G_1 G_3$  (D)  $2(A_1 + A_2)G_1^2 G_3^2$

**Ans.** C

**Sol.** Let the two numbers are  $a, b$ .

$$A_1 = a + \frac{b-a}{3} = \frac{2a+b}{3}$$

$$A_2 = a + \frac{b-a}{3} \cdot 2 = \frac{a+2b}{3}$$

$$G_1 = a \left( \frac{b}{a} \right)^{\frac{1}{4}}$$

$$G_2 = a \left( \frac{b}{a} \right)^{\frac{2}{4}}$$

$$G_3 = a \left( \frac{b}{a} \right)^{\frac{3}{4}}$$

$$(G_1)^4 + (G_2)^4 + (G_3)^4 + (G_1)^2 \cdot (G_3)^2$$

$$\Rightarrow a^4 \cdot \frac{b}{a} + a^4 \cdot \frac{b^2}{a^2} + a^4 \cdot \frac{b^3}{a^3} + a^4 \cdot \frac{b^2}{a^2}$$

$$= ba^3 + b^2 a^2 + b^3 a + a^2 b^2$$

$$= ab(a^2 + b^2 + 2ab) = ab(a+b)^2$$

$$(A_1 + A_2)^2 \cdot G_1 G_3 = (a+b)^2 \cdot ab$$

**69.** The sum of first four terms of a geometric progression (G.P.) is  $\frac{65}{12}$  and the sum of their respective reciprocals is

$\frac{65}{18}$ . If the product of first three terms of the G.P. is 1, and the third term is  $\alpha$ , then  $2\alpha$  is \_\_\_\_.

एक गुणोत्तर श्रेणी के पहले चार पदों का योग  $\frac{65}{12}$  है तथा उनके व्युत्क्रमों का योग  $\frac{65}{18}$  है। यदि इसके पहले तीन पदों का गुणनफल 1 हो और तीसरा पद  $\alpha$  हो, तो  $2\alpha$  बराबर है \_\_\_\_\_।

Ans. C

Sol. GP  $a, ar, ar^2, ar^3$

$$a + ar + ar^2 + ar^3 = \frac{65}{12}$$

$$S_4 = a \frac{(r^4 - 1)}{r - 1} = \frac{65}{12} \quad \dots\dots\dots(1)$$

$$\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \frac{1}{ar^3} = \frac{65}{18}$$

$$\frac{\frac{1}{ar^3}(r^4 - 1)}{r - 1} = \frac{65}{18} \quad \dots\dots\dots(2)$$

$$(A) / (2)$$

$$a^2 r^3 = \frac{3}{2}$$

$$a \times ar \times ar^2 = 1$$

$$a^3 r^3 = 1$$

$$\Rightarrow a = \frac{2}{3}$$

$$r = \frac{3}{2}$$

$$\alpha = ar^2 = \frac{3}{2}$$

$$2\alpha = 3$$

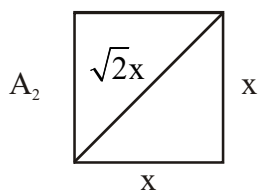
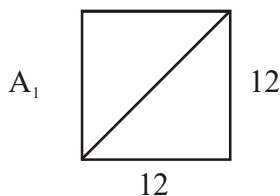
70. Let  $A_1, A_2, A_3, \dots$  be squares such that for each  $n \geq 1$ , the length of the side of  $A_n$  equals the length of diagonal of  $A_{n+1}$ . If the length of  $A_1$  is 12 cm, then the smallest value of  $n$  for which area of  $A_n$  is less than one, is \_\_\_\_\_.

माना  $A_1, A_2, A_3, \dots$  वर्ग हैं जबकि प्रत्येक  $n \geq 1$  के लिए  $A_n$  की भुजा की लम्बाई  $A_{n+1}$  के विकर्ण की लम्बाई के बराबर

है। यदि  $A_1$  की भुजा की लम्बाई 12 cm है, तो  $n$  का न्यूनतम मान, जिसके लिए  $A_n$  का क्षेत्रफल एक से कम है, है :

Ans. 9

Sol.



given,

$$\sqrt{2}x = 12$$

$$x = 6\sqrt{2}$$

| Square | Side                 | Area               |
|--------|----------------------|--------------------|
| $A_1$  | 12                   | 144                |
| $A_2$  | $6\sqrt{2}$          | 72                 |
| —      | —                    | —                  |
| —      | —                    | —                  |
| —      | —                    | —                  |
| $A_8$  | $\frac{3}{\sqrt{2}}$ | $\frac{9}{8}$      |
| $A_9$  | $\frac{3}{4}$        | $\frac{9}{16} < 1$ |

$$n = 9$$

71. The total number of two digit numbers 'n', such that  $3^n + 7^n$  is a multiple of 10, is \_\_\_\_\_ .

दो अंको की संख्याओं 'n', जिनके लिए  $3^n + 7^n$ , 10 का गुणज है, की कुल संख्या है \_\_\_\_\_ ।

Ans. 45

Sol.  $3^n + 7^n$  is divisible by 3 + 7 iff n is odd

11, 13, 15, ....., 99

number of two digit numbers 'n' = 45

72. Let  $S_n(x) = \log_{a^{1/2}} x + \log_{a^{1/3}} x + \log_{a^{1/6}} x + \log_{a^{1/11}} x + \log_{a^{1/18}} x + \log_{a^{1/27}} x + \dots$  up to n-terms, where  $a > 1$ .

If  $S_{24}(x) = 1093$  and  $S_{12}(2x) = 265$ , then value of  $a$  is equal to \_\_\_\_\_.

माना  $S_n(x) = \log_{a/2} x + \log_{a/3} x + \log_{a/6} x + \log_{a/11} x + \log_{a/18} x + \log_{a/27} x + \dots n$  पदों तक, जहाँ  $a > 1$  है।

यदि  $S_{24}(x) = 1093$  है। तथा  $S_{12}(2x) = 265$  हैं, तो  $a$  का मान बराबर है \_\_\_\_\_।

Ans. 16

Sol.  $S_n(x) = 2\log_a x + 3\log_a x + 6\log_a x + 11\log_a x + \dots n$  terms

$$S_n(x) = (2 + 3 + 6 + 11 + \dots + n \text{ terms}) \log_a x$$

$$\text{Let, } S = 2 + 3 + 6 + 11 + \dots + T_n$$

$$S = 2 + 3 + 6 + \dots + T_{n-1} + T_n$$

$$0 = 2 + (1 + 3 + 5 + \dots + (n-1) \text{ terms}) + T_n$$

$$T_n = 2 + (n-1)^2 = n^2 - 2n + 3$$

$$S = \Sigma(n^2 - 2n + 3) = \frac{n}{6} [2n^2 - 3n + 13]$$

$$S_n(x) = \frac{n}{6} (2n^2 - 3n + 13) \log_a x$$

$$S_{24}(x) = 4 \times 1093 \log_a x = 1093$$

$$\log_a x = \frac{1}{4}$$

$$S_{12}(2x) = 2 \times 265 (\log_a 2 + \log_a x) = 265$$

$$2 \left( \log_a 2 + \frac{1}{4} \right) = 1$$

$$\log_a 2 = \frac{1}{4}$$

$$a = 2^4 = 16$$

73. Let  $\frac{1}{16}, a$  and  $b$  be in G.P. and  $\frac{1}{a}, \frac{1}{b}, 6$  be in A.P. in A.P. where  $a, b > 0$ . then  $72(a + b)$  is equal to \_\_\_\_\_.

माना  $\frac{1}{16}, a$  तथा  $b$  G.P. में है तथा  $\frac{1}{a}, \frac{1}{b}, 6$  A.P. में है, जहाँ  $a, b > 0$  हैं, तो  $72(a + b)$  बराबर है \_\_\_\_\_।

Ans. 14

Sol.  $\frac{1}{16}, a, b$  are in G.P.

$$a^2 = \frac{b}{16} \Rightarrow b = 16a^2 \dots (i)$$

$$\frac{1}{a}, \frac{1}{b}, 6 \text{ are in A.P.}$$

$$\frac{2}{b} = \frac{1}{a} + 6$$

$$\frac{2}{16a^2} = \frac{1}{a} + 6$$

$$48a^2 + 8a - 1 = 0$$

$$a = \frac{1}{12} \text{ or } \frac{-1}{4} (a > 0)$$

$$a = \frac{1}{12}, b = \frac{1}{9}$$

$$72(a + b) = 14$$

74. Consider an arithmetic series and a geometric series having four initial terms from the set {11, 8, 21, 16, 26, 32, 4}. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to \_\_\_\_\_.

एक समान्तर श्रेणी तथा एक गुणोत्तर श्रेणी के पहले चार पद समुच्चय {11, 8, 21, 16, 26, 32, 4} में से हैं। यदि इन श्रेणियों के अंतिम पद चार अंकों की अधिकतम संभव संख्याएँ हैं, तो दोनों श्रेणियों में होने वाले पदों की संख्या है \_\_\_\_\_।

Ans. C

Sol. G.P.  $\equiv \{4, 8, 16, 32, \dots\}$

A.P.  $\equiv \{11, 16, 21, 26, \dots\}$

G.P. will end at 8192

All terms of GP whose units digit is 6 will also be part of A.P.

Common terms  $\equiv \{16, 256, 4096\}$

Ans = 3

75. The 4<sup>th</sup> term of GP is 500 and its common ratio is  $\frac{1}{m}$ ,  $m \in \mathbb{N}$ . Let  $S_n$  denote the sum of the first  $n$  terms of this GP.

If  $S_6 > S_5 + 1$  and  $S_7 < S_6 + \frac{1}{2}$ , then the number of possible values of  $m$  is \_\_\_\_\_.

एक GP का चौथा पद 500 है तथा इसका सार्वअनुपात  $\frac{1}{m}$ ,  $m \in \mathbb{N}$  है। माना इस GP के प्रथम  $n$  पदों का योग  $S_n$  है। यदि  $S_6 > S_5 + 1$  और  $S_7 < S_6 + \frac{1}{2}$ , तो संभव संख्याएँ  $m$  की हैं \_\_\_\_\_।

+ 1 तथा  $S_7 < S_6 + \frac{1}{2}$  हो, तो  $m$  के संभव मानों की संख्या है।

**Ans.** 12

**Sol.** Given 4<sup>th</sup> term of GP = 500

$$\text{C.R.} = \frac{1}{m}$$

$$\frac{a}{m^3} = 500$$

$$S_n = \frac{a \left[ \left( \frac{1}{m} \right)^n - 1 \right]}{\frac{1}{m} - 1}$$

$$S_n = \frac{a(1 - m^n)}{m^n(1 - m)}$$

$$S_n = \frac{a(1 - m^n)}{m^{n-1}(1 - m)}$$

$$S_6 > S_5 + 1$$

$$\frac{a(1 - m^6)}{m^5(1 - m)} > \frac{a(1 - m^5)}{m^4(1 - m)} + 1$$

$$a - am^6 > m^5 - m^6 + am - am^6$$

$$a(1 - m) > m^5(1 - m)$$

$$a > m^5 \quad \dots(2)$$

$$S_7 < S_6 + \frac{1}{2}$$

$$\frac{a(1 - m^7)}{m^6(1 - m)} < \frac{a(1 - m^6)}{m^5(1 - m)} + \frac{1}{2}$$

$$a - am^7 < am - am^7 + \frac{1}{2}m^6 - \frac{m^7}{2}$$

$$a(1 - m) < \frac{1}{2}m^6(1 - m)$$

$$a < \frac{m^6}{2} \dots(3)$$

$$\text{So } m^5 < a < \frac{m^6}{2}$$

$$m^2 < \frac{a}{m^3} < \frac{m^3}{2}$$

$$m^2 < 500 < \frac{m^3}{2}$$

$$m \in [11, 22]$$

so, no. of possible value of m is 12

76. If  $\frac{1^3 + 2^3 + 3^3 + \dots \text{up to } n \text{ term}}{1.3 + 2.5 + 3.7 + \dots \text{up to } n \text{ term}} = \frac{9}{5}$ , then the value of n is \_\_\_\_\_.

यदि  $\frac{1^3 + 2^3 + 3^3 + \dots \text{ n terms}}{1.3 + 2.5 + 3.7 + \dots \text{ n terms}} = \frac{9}{5}$ , है तो n का मान है।

Ans. 5

Sol.  $\frac{1^3 + 2^3 + 3^3 + \dots \text{upto } n \text{ terms}}{1.3 + 2.5 + 3.7 + \dots \text{upto } n \text{ terms}}$

$$\Rightarrow \frac{\frac{n^2(n+1)^2}{4}}{\Sigma n(2n+1)} = \frac{9}{5}$$

$$\frac{\frac{n^2(n+1)^2}{4}}{2 \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}} = \frac{9}{5}$$

$$\frac{\frac{n(n+1)}{4} \cdot 6}{2(2n+1) + 3} = \frac{9}{5}$$

$$5n^2 - 19n - 30 = 0$$

$$n = 5, -\frac{6}{5}$$

$$n = 5$$

77. Let  $A_1, A_2, A_3$ , be the three A.P. with the same common difference d and having their first terms as  $A, A + 1, A + 2$ , respectively. Let a, b, c be the 7<sup>th</sup>, 9<sup>th</sup>, 17<sup>th</sup> terms of  $A_1, A_2, A_3$ , respectively such that

$$\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0. \text{ If } a = 29, \text{ then the sum of first 20 terms of an AP whose first term is } c-a-b \text{ and common}$$

difference is  $\frac{d}{12}$ , is equal to \_\_\_\_\_.

माना  $A_1, A_2, A_3$  तीन A.P. है, जिनका सार्वअंतर  $d$  है तथा जिनके पहले पद क्रमशः  $A, A+1, A+2$ , है। माना  $A_1, A_2, A_3$ , के 7वाँ,

$$9वाँ, 17वाँ पद क्रमशः a, b, c हैं तथा \begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0 \text{ है। यदि } a = 29 \text{ है, तो उस AP जिसका पहला पद } c-a-b \text{ है तथा}$$

सार्वअंतर  $\frac{d}{12}$  है, के प्रथम 20 पदों का योग बराबर है।

**Ans.** 495

**Sol.**  $A + 6d = a$  \_\_\_\_\_ (1)

$$A + 1 + 8d = b$$
 \_\_\_\_\_ (2)

$$A + 2 + 16d = c$$
 \_\_\_\_\_ (3)

Now

$$\begin{vmatrix} a & 7 & 1 \\ 2b & 17 & 1 \\ c & 17 & 1 \end{vmatrix} + 70 = 0$$

$$\Rightarrow c - 2b = 7$$

Using equation (2) & (3), we get  $A = -7$

$$\text{Now } a = -7 + 6d = 29 \Rightarrow d = 6$$

Now first term =  $c-a-b = 20$  and common difference =  $\frac{d}{12} = 1/2$  and the sum of first 20 terms of an

$$AP = 495$$

- 78.** For the two positive numbers  $a, b$ , if  $a, b$  and  $\frac{1}{18}$  are in geometric progression, while  $\frac{1}{a}, 10$  and  $\frac{1}{b}$  are in an arithmetic progression, then  $16a + 12b$  is equal to \_\_\_\_\_.

दो घनात्मक संख्याओं  $a, b$  के लिए, यदि  $a, b$  तथा  $\frac{1}{18}$  एक गुणोत्तर श्रेढ़ी में है और  $\frac{1}{a}, 10$  तथा  $\frac{1}{b}$  एक समांतर श्रेढ़ी में है, तो

$$16a + 12b \text{ बराबर है।}$$

**Ans.** C

**Sol.**  $a, b, \frac{1}{18} \rightarrow GP$

$$\frac{a}{18} = b^2 \quad \dots(i)$$

$\frac{1}{a}, 10, \frac{1}{b} \rightarrow AP$

$$\frac{1}{a} + \frac{1}{b} = 20$$

$\Rightarrow a + b = 20ab$ , from eq. (i) ; we get

$$\Rightarrow 18b^2 + b = 360b^3$$

$$\Rightarrow 360b^2 - 18b - 1 = 0 \quad \{\because b \neq 0\}$$

$$\Rightarrow b = \frac{18 \pm \sqrt{324 + 1440}}{720}$$

$$\Rightarrow b = \frac{18 + \sqrt{1764}}{720} \quad \{\because b > 0\}$$

$$\Rightarrow b = \frac{1}{12}$$

$$\Rightarrow a = 18 \times \frac{1}{144} = \frac{1}{8}$$

$$\text{Now, } 16a + 12b = 16 \times \frac{1}{8} + 12 \times \frac{1}{12} = 3$$

**79.** Let  $a_1, a_2, a_3, \dots$  be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then  $a_1a_9 + a_2a_4a_9 + a_5 + a_7$  is equal to \_\_\_\_\_ .

माना  $a_1, a_2, a_3, \dots$  वर्धमान धनात्मक संख्याओं की एक GP है। यदि चौथे व छठवें पदों का गुणनफल 9 है और पाँचवें व सातवें पदों का योग 24 है, तब  $a_1a_9 + a_2a_4a_9 + a_5 + a_7$  बराबर है।

**Ans.** 60

**Sol.**  $\because a_1, a_2, a_3, \dots$  increasing G.P. of positive nos.

$$\because a_4a_6 = 9 \quad \text{and } a_5 + a_7 = 24$$

$$(ar^3)(ar^5) = 9$$

$$a^2r^8 = 9 \quad \text{and } ar^4 + ar^6 = 24$$

$$\Rightarrow ar^4 = 3 \quad \dots(1) \quad ar^4(1 + r^2) = 24 \quad \dots(2)$$

$$\therefore \text{ from (1) and (2) we get } 1 + r^2 = 8$$

$$r^2 = 7 \Rightarrow r = \sqrt{7}$$

$$\therefore a = \frac{3}{49}$$

$$\therefore a_1 \cdot a_9 + a_2 \cdot a_4 a_9 + a_5 + a_7$$

$$= a(ar^8) + (ar)(ar^3)(ar^8) + ar^4 + ar^6$$

$$= a^2r^8 + a^3r^{12} + ar^4 + ar^6$$

$$= 9 + 3^3 + 3 + \frac{3}{49} \times 7^3$$

$$= 9 + 27 + 3 + 21$$

$$= 60$$

**80.** Let  $\{a_k\}$  and  $\{b_k\}$ ,  $k \in \mathbb{N}$ , be two G.P.s with common ratios  $r_1$  and  $r_2$  respectively such that  $a_1 = b_1 = 4$  and

$r_1 < r_2$ . Let  $c_k = a_k + b_k$ ,  $k \in \mathbb{N}$ . If  $c_2 = 5$  and  $c_3 = \frac{13}{4}$  then  $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$  is equal to \_\_\_\_\_.

माना  $\{a_k\}$  तथा  $\{b_k\}$ ,  $k \in \mathbb{N}$ , दो G.P.s हैं, जिनके सार्व अनुपात क्रमशः  $r_1$  तथा  $r_2$  हैं, और  $a_1 = b_1 = 4$ ,  $r_1 < r_2$  है। माना

$c_k = a_k + b_k$ ,  $k \in \mathbb{N}$  है। यदि  $c_2 = 5$  तथा  $c_3 = \frac{13}{4}$  है, तो  $\sum_{k=1}^{\infty} c_k - (12a_6 + 8b_4)$  बराबर है।

**Ans.** 9

**Sol.**  $a = b = 4$

$$C_2 = a_2 + b_2 = 5$$

$$\Rightarrow ar_1 + br_2 = 5$$

$$\Rightarrow 4r_1 + 4r_2 = 5 \Rightarrow r_1 + r_2 = \frac{5}{4}$$

$$C_3 = a_3 + b_3 = \frac{13}{4} \Rightarrow ar_1^2 + br_2^2 = \frac{13}{4}$$

$$\Rightarrow 4r_1^2 + 4r_2^2 = \frac{13}{4}$$

$$\Rightarrow 4\left(\frac{5}{4} - r_2\right)^2 + 4r_2^2 = \frac{13}{4}$$

$$\Rightarrow 16\left(\frac{25}{16} - \frac{5}{2}r_2 + r_2^2\right) + 16r_2^2 = 13$$

$$\Rightarrow 25 - 40r_2 + 16r_2^2 + 16r_2^2 - 13 = 0$$

$$\Rightarrow 32r_2^2 - 40r_2 + 12 = 0$$

$$\Rightarrow 8r_2^2 - 10r_2 + 3 = 0$$

$$\Rightarrow (4r_2 - 3)(2r_2 - 1) = 0$$

$$\Rightarrow r_2 = \frac{3}{4} \text{ or } \frac{1}{2}$$

$$r_1 = \frac{1}{2} \text{ or } \frac{3}{4}$$

$$r_1 < r_2 \Rightarrow r_1 = \frac{1}{2}, r_2 = \frac{3}{4}$$

$$\sum_{k=1}^{\infty} C_k = \sum a_k + \sum b_k = \frac{a}{1-r_1} + \frac{b}{1-r_2} = \frac{4}{1-\frac{1}{2}} + \frac{4}{1-\frac{3}{4}}$$

$$= 8 + 16 = 24$$

$$\sum_{k=1}^{\infty} C_k - 12a_6 - 8b_4 = 24 - 12 \left( 4 \left( \frac{1}{2} \right)^5 \right) - 8 \cdot 4 \cdot \left( \frac{3}{4} \right)^3$$

$$= 24 - \frac{3}{2} - \frac{27}{2} = 9$$